Cellular Respiration Presentation Script

Unit 3B Slides 1-7

Slide 1: Cellular Respiration

We have already established that ATP is the cell's immediate source of usable energy. The question now is how cells continually regenerate ATP from ADP and phosphate. Cellular respiration is the set of metabolic pathways that transfers energy stored in food molecules, especially glucose, into ATP. This is not the same as breathing. Breathing brings oxygen into the body and removes carbon dioxide, while cellular respiration is a chemical process occurring within cells.

It is also important to avoid thinking of respiration as one explosive reaction. If a cell released all of glucose's energy at once, it would lose much of that energy as heat and could damage the cell. Instead, enzymes control a sequence of smaller reactions. At each stage, the cell captures a portion of the released energy in ATP or in electron carriers that will be used later.

Slide 2: Cellular Respiration Overview

The central goal of cellular respiration is to convert the chemical energy stored in organic molecules into ATP. Glucose is our model fuel because carbohydrates are usually accessed first, but cells can also break down lipids and proteins. Those molecules enter the respiratory pathways at different points.

The overall equation is: glucose plus oxygen produces carbon dioxide, water, and released energy. In balanced form, one glucose reacts with six oxygen molecules to form six carbon dioxide molecules and six water molecules. Some of the released energy is captured in ATP, while some is released as heat. ATP is often written above the reaction arrow or beside the equation to show the energy outcome. Still, it is not usually treated as a matter product like carbon dioxide and water.

Notice that the atoms are conserved. The carbon atoms from glucose eventually leave as carbon dioxide. Oxygen atoms appear in both carbon dioxide and water. The equation summarizes many enzyme-controlled reactions, so it does not show the individual electron and energy transfers that make ATP production possible.

Slide 3: Mitochondrial Structures

In eukaryotic cells, mitochondria carry out most aerobic respiration. Structure and function are closely related here. The outer membrane forms the organelle's boundary, while the inner membrane is highly folded. These folds, called cristae, greatly increase the membrane surface area available for proteins that carry out the electron transport chain and produce ATP. More surface area allows more of these protein complexes to operate at the same time.

The intermembrane space lies between the outer and inner membranes. Later, hydrogen ions will accumulate in this compartment, creating an electrochemical gradient. The matrix is the fluid-filled space enclosed by the inner membrane. Pyruvate oxidation and the Krebs cycle occur there. When labeling the diagram in your notes, distinguish the inner membrane itself from the cristae, which are folds of that membrane.

Mitochondria also contain their own DNA and ribosomes. This evidence, along with their double membrane and ability to divide, supports the endosymbiotic theory: ancestral mitochondria were once free-living prokaryotic cells that entered into a mutually beneficial relationship with a larger host cell.

Slide 4: Lipid Bilayers

Mitochondrial membranes are phospholipid bilayers, like the cell membrane. Their hydrophilic phosphate heads face the watery environments, and their hydrophobic fatty-acid tails face inward. This arrangement creates a selectively permeable barrier. Charged hydrogen ions cannot freely cross the bilayer's hydrophobic interior.

That property becomes essential during the electron transport chain. Proteins in the inner membrane pump hydrogen ions from the matrix into the intermembrane space. Because the lipid bilayer restricts their movement, the ions accumulate on one side. Their concentration and electrical differences store potential energy, much like water held behind a dam. Hydrogen ions can return to the matrix primarily through ATP synthase, and that controlled flow provides energy for ATP production.

This shows how compartmentalization increases cellular efficiency. The inner membrane separates two chemically different environments and lets the cell control ion movement. If the inner membrane became freely permeable to hydrogen ions, the gradient would dissipate, and ATP production would decrease sharply.

Slide 5: Four Stages of Cellular Respiration

Cellular respiration has four major stages. First, glycolysis occurs in the cytosol and splits one six-carbon glucose into two three-carbon pyruvate molecules. Glycolysis does not directly require oxygen, so it is classified as anaerobic. It produces a small net gain of ATP and transfers electrons to NADH.

If oxygen is available, the two pyruvate molecules enter the mitochondrion. During pyruvate oxidation, each pyruvate is converted to acetyl-CoA. The acetyl groups then enter the Krebs cycle in the matrix. These middle stages remove carbon as carbon dioxide and transfer high-energy electrons to NADH and FADH2. The fourth stage is the electron transport chain, located in the inner mitochondrial membrane. This stage uses the stored electron energy to establish a hydrogen-ion gradient and produce most of the ATP.

The terms anaerobic and aerobic describe whether oxygen is used. Glycolysis is anaerobic and occurs in the cytosol. The later stages are considered aerobic because they require oxygen as the final electron acceptor, even though oxygen is used directly only at the end of the electron transport chain. In this course packet, anaerobic respiration refers to glycolysis followed by fermentation when oxygen is unavailable.

Slide 6: Overview of Cellular Respiration

As we trace the overview diagram, follow the carbon atoms first. Glucose begins with six carbons. Glycolysis rearranges and splits it into two three-carbon pyruvate molecules. Pyruvate oxidation removes one carbon from each pyruvate as carbon dioxide, leaving two-carbon acetyl groups. The Krebs cycle completes the removal of those carbons as additional carbon dioxide. This explains why carbon dioxide is a waste product of glucose oxidation.

Next, follow the energy. A small amount of ATP is made directly during glycolysis and the Krebs cycle through substrate-level phosphorylation. However, much of the energy is temporarily stored in NADH and FADH2. These carriers deliver high-energy electrons to the electron transport chain. The ETC uses that energy to build the hydrogen-ion gradient that drives ATP synthase, so most ATP is produced during oxidative phosphorylation.

Finally, follow oxygen. Oxygen does not enter glycolysis or the Krebs cycle. It is used at the end of the electron transport chain, where it accepts electrons and combines with hydrogen ions to form water. Without oxygen, the electron transport chain backs up, the carriers cannot unload their electrons efficiently, and oxidative phosphorylation stops.

Slide 7: Gradual Oxidation and Electron Carriers

Glucose is gradually oxidized rather than broken down in a single step. Oxidation means losing electrons, often along with hydrogen. Reduction means gaining electrons or hydrogen. The memory aid OIL RIG reminds us that Oxidation Is Loss and Reduction Is Gain. When glucose and its intermediate molecules are oxidized, their electrons still contain usable energy.

NAD+ and FAD are coenzymes that carry electrons. When NAD+ gains high-energy electrons and hydrogen, it becomes NADH. When FAD is reduced, it becomes FADH2. The reduced forms carry electrons from glycolysis, pyruvate oxidation, and the Krebs cycle to the electron transport chain. There, the carriers are oxidized as they donate their electrons, returning to NAD+ and FAD so they can be reused.

A useful way to think about these molecules is as rechargeable electron shuttles. NAD+ and FAD are the unloaded forms; NADH and FADH2 are the loaded forms. The carriers do not create energy. The cell transfers energy from glucose to the carriers and later uses it in the electron transport chain. This controlled series of transfers allows the cell to capture substantially more energy than it could through glycolysis alone.

Closing Statement

Cellular respiration is best understood as an organized transfer of matter and energy. Carbon from glucose leaves as carbon dioxide, electrons move through NADH and FADH2, oxygen accepts the electrons at the end of the pathway, and the inner mitochondrial membrane uses an ion gradient to produce ATP. In the next part of the lesson, we will examine glycolysis as the first stage of this sequence.